🐢Turtle Graphics Final Project

Instructions & Rubric

🌟 Project Theme

Create a **Winter or Christmas–themed artwork** using Python Turtle.
Your artwork should be creative, colorful, and complete.

📅Project Deadline

12.19th

📘 Project Requirements

To receive full credit, your Turtle program must include all the following components:

---

1. Functions（20 pts）

You must use **at least 2 custom functions**.

Your functions must meet the following criteria:

(1) Correctly define functions

Example:

```
def draw_tree():
    # your code

```

(2) Correctly call functions

Example:

```
draw_tree()

```

(3) Functions must be meaningful

- Good: A function draws one complete object (tree, snowflake, house…).
- Not good: A function only draws *one line* or *one circle* repeatedly used only once.

(4) Function names must be clear and readable

✅ Correct examples (good naming)

- draw_tree()

- draw_snowflake()
- draw_gift_box()
- draw_snowman()

❌ **Incorrect examples (bad naming)**

- d()
- function1()
- aaa()
- test()

2. For Loops（20 pts）

Your program must include at least **one meaningful for-loop** (looping 3 or more times).

✔ Correct use example

- Draw repeating patterns
- Draw multiple snowflakes
- Draw decorations on a tree
- Draw steps of a snowman's buttons

✔Loop must be meaningful

✅Good loop:

```
for i in range(6):
    draw_snowflake()
```

❌Not good loop:

```
for i in range(1):
    draw_tree()
```

(looping only once = meaningless)

3. Color & Fill（25 pts）

✔ (1) Must use bgcolor()

The background **must** be set, even if you want to use a white background.

✔ (2) The drawing must NOT be only black and white

At least 3 colors should be used in the drawing.

✔ (3) Must use filled shapes

You must use both:

```
begin_fill()
...
end_fill()
```

At least **one object** must be filled with color.

4. Write Your Name and Class（10 pts）

- Must be **clearly visible**
- Placed at a **reasonable location** (bottom corner recommended)
- Must include both **name + class**

Example:

```
write("Edward Wang G7-5", font=("Arial", 16, "bold"))

```

🎨 Creativity & Design（15 pts）

Evaluation includes:

- Visual completeness
- Balance of elements
- Clear winter/holiday theme
  (Snowflakes / trees / snowman / gifts / stars / winter houses / ornaments / holiday lights, etc.)

🧹 Code Organization & Cleanliness（10 pts）

- Code is tidy and readable
- Functions are grouped neatly
- No long copy-paste repeated code
- Good use of spacing and comments

🚫 Points Deducted for Missing Requirements

Missing Requirement	Deduction
No functions	–20 pts
Function name unclear	–3 to –5 pts per function
Functions exist but meaningless	–10 pts
No for-loop	–20 pts
Loop meaningless (range(1))	–10 pts
No background color	–5 pts
Only black/white drawing	–10 pts
No filled shapes	–10 pts
No name or class on screen	–10 pts
Name too small/not visible	–3 pts
Theme unrelated to winter/Christmas	–10 pts
Late Submssion	Base on late date

📊 Total Score: 100 pts

- Functions: 20
- For Loops: 20
- Color & Fill: 25
- Name & Class: 10
- Creativity: 15
- Code Cleanliness: 10
  Total = 100